

Designing 8-to-1 Multiplexer Using LabVIEW



DR M. SATYANARAYANA AND
M.A.K. MUZEBUDDIN

A multiplexer or MUX is a device that selects one of several analogue or digital input signals and forwards the selected input into a single line at the output. Multiplexers are mainly used to increase the amount of data that can be sent over the network within a certain amount of time and bandwidth. A multiplexer is also called a data selector.

This article explains the step-by-step procedure to design an 8-to-1 MUX using National Instruments' LabVIEW program. A multiplexer of 2^n inputs has n select lines, which are used to select which input line needs to be sent to the output. In this design, we have eight input lines and three select lines.

First install LabVIEW in your system. We used both LabVIEW 2013 and 2011 versions and found the program working well. The evaluation version of LabVIEW is available for free download from the <http://www.ni.com/trylabview/>

Steps for designing

We will use Controls palettes of LabVIEW in our design. The Controls palettes contain the front panel controls and indicators for designing the user-interface program. Note, LabVIEW programs/subroutines are called virtual instruments (VIs).

1. Launch the LabVIEW from the desktop and then select New VI option from the File menu, or press Ctrl+N keys from the keyboard followed by pressing Ctrl+T to open a new VI project. You will see a new front panel on the screen.

2. Right click on the front panel and select Modern → Numeric → Numeric Control (this is where we give inputs).

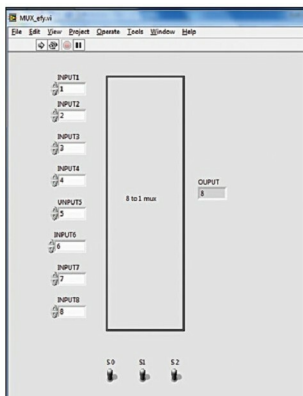


Fig. 1: Front panel window

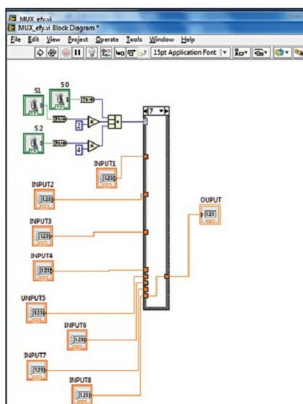


Fig. 2: Block diagram of 8-to-1 MUX wire connections

Drag and drop it on to the front panel. Label the control input as input 1. Similarly, select other seven such inputs and label them as input 2,... input 8, respectively. Arrange the inputs in the sequential order.

3. Next, select Modern → Decorations → Flat Frame and adjust its size such that all the inputs are positioned on its left side for a neat appearance (refer Fig. 1). Double left click on the front panel and type '8 to 1 Mux' and drag it inside the frame box.

4. Once again, go to Modern → Numeric → Numeric Indicator and drag it on to the front panel. This is where we see the output. So this indicator is placed on the right side of the frame.

5. Now, click on Modern → Boolean → Vertical Toggle switch, place it below frame box and label it as S0. Select two more vertical toggle switches, arrange them below the box and label them S1 and S2. This completes the user-interface design on the front panel. After building the user interface, you can add the code in the block diagram using VIs and structures to control the front panel objects. Select Window menu option, click on Show Block Diagram to view the block diagram.

6. Next, right click on the block diagram, select Programming → Structures → Case Structure, drag and drop it on to the block diagram. You will find a menu indicator box on top of the Case Structure. If